

Unit V

Q. 1 Explain the concept of chemical equilibrium with characteristics.

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For a general reversible reaction:



• At equilibrium:

- The rate of forward reaction

$$r_1 = k_1 [A]^a [B]^b$$

- The rate of reverse reaction

$$r_2 = k_2 [C]^c [D]^d$$

- At equilibrium

$$r_1 = r_2$$

• equilibrium constant.

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

★ characteristics of chemical equilibrium.

a. Dynamic in nature

- although the macroscopic properties remain constant, molecular-level forward and reverse reactions continue to occur at equal rates.

b. attained in closed system

- equilibrium is established in a closed system where no reactants & products are added or removed.

c. Constant concentrations.

- at equilibrium, the concentration of reactant and product remain constant over time.

d. dependence on initial conditions.

- initial concentration

- temp

- pressure

- catalyst.

e. temp dependence.

According to Le Chatelier's principle equilibrium shifts with temp. changes.

- Exothermic reaction: increases in temp shift equilibrium left.
- endothermic reaction: increases shift it right.

f. not indication of completion.

- equilibrium does not mean 100% conversion.

2] Explain heat of reaction from thermodynamics

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The heat liberated or absorbed during reaction at temperature T_2 depends on the nature of the reacting system, the amount of material reacting, and the temp and pressure of the reacting system, and is calculated from the heat of reaction ΔH_r .

- When this is not known, it can be calculated from the heats of formation ΔH_f or heat of combustion ΔH_c of reacting material.

Consider a reaction $aA \rightarrow rR + sS$.

By convention we define the heat of reaction at temp T as the heat transferred to the reacting system from the surrounding when a moles of A disappears to produce r moles of R and s moles of S with the system measured at the same temp and pressure before and after the change.

$\therefore aA \rightarrow rR + sS$, $\Delta H_{rT} \begin{cases} +ve, \text{ endothermic} \\ -ve, \text{ exothermic} \end{cases}$

To evaluate the heat of reaction at T_2 knowing the heat of reaction at T_1 .

∴ By law of conservation of energy.

$$\left(\begin{array}{l} \text{heat absorbed} \\ \text{during rxn at} \\ \text{temp } T_2 \end{array} \right) = \left(\begin{array}{l} \text{heat added to} \\ \text{reactants to} \\ \text{change their} \\ \text{temp from } T_2 \text{ to } T_1 \end{array} \right) + \left(\begin{array}{l} \text{heat absorbed} \\ \text{during rxn} \\ \text{at temp } T_1 \end{array} \right) + \left(\begin{array}{l} \text{heat} \\ \text{produced} \\ \text{them} \\ \text{to } T_2 \end{array} \right)$$

In terms of enthalpies of reactant and product

$$\Delta H_{r2} = -(H_2 - H_1)_{\text{reactant}} + \Delta H_{r1} + (H_2 - H_1)_{\text{product}}$$

$$\Delta H_{r2} = \Delta H_{r1} + \int_{T_1}^{T_2} \Delta C_p dT$$

$$\Delta C_p = C_{pR} + \sum C_{pS} - \sum C_{pA}$$

$$\text{where } C_{pA} = \alpha_A + \beta_A T + \gamma_A T^2$$

$$C_{pR} = \alpha_R + \beta_R T + \gamma_R T^2$$

$$C_{pS} = \alpha_S + \beta_S T + \gamma_S T^2$$

$$\Delta H_{r2} = \Delta H_{r1} + \int_{T_1}^{T_2} (\sum \alpha + \sum \beta T + \sum \gamma T^2) dT$$

Q.3) Explain optimum temp. progression for exothermic reversible reaction.

→ The otp is temp profile that maximize the reaction rate and conversion. Since the rxn is exothermic, heat is released and the equilibrium conversion decreases with increase in temp.

• OTP characteristics.

1) Initial high temp: Start with a higher temp to maximize the rxn rate.

2) Gradual decrease in temp: As the reaction

proceeds and conversion increase, decrease the temp to favor equilibrium conversion.

• Application:

The OTP is crucial in designing and operating reactors for exothermic reversible reaction such as:

- ① Ammonia synthesis: $N_2 + 3H_2 \rightleftharpoons 2NH_3$
- ② methanol synthesis: $CO + 2H_2 \rightleftharpoons CH_3OH$

8.4) Explain effect of temp on heat of rxn.

→ In CRE, the heat of reaction is the amount of heat released or absorbed during chemical reaction. This value is not constant and can change with time temp.

- When the rxn is exothermic it releases heat and when it is endothermic it absorbs heat. As the temp changes, the heat capacities (C_p) of the reactant and product also affect the overall heat of rxn.

- According to Kirchhoff's law, the heat of reaction at different temp can be calculated by considering the diff in heat capacities of product and reactant.

- If the heat capacities of the product are higher than those of reactants the heat of reaction become less exothermic as temp increase & if the heat capacities of reactant are higher than those of product the heat of rxn become more exothermic with temp.

- This change in heat of rxn with temp is important for designing reactors

Q. Explain non-adiabatic operations,

→ In industrial practice, the re^n temp are controlled by either addition or removal of heat depending upon the situation. This is required because of demands from kinetic and thermodynamic consideration. When we provide such type of temp. control, the re^n become non-adiabatic.

- Let Q be the total heat added to a reactor per mole of reactant A entering the reactor.

- The energy balance about the system yields,

$$Q = C_p'' (T_2 - T_1) x_A + C_p' (T_2 - T_1) (1 - x_A) + \Delta H_{R1} x_A \quad \text{---(1)}$$

$$Q = C_p'' (T_2 - T_1) x_A + (C_p' (T_2 - T_1) - C_p' (T_2 - T_1) x_A) + \Delta H_{R1} x_A$$

$$- x_A [\Delta H_{R1} + C_p'' (T_2 - T_1) - C_p' (T_2 - T_1)] = (C_p' (T_2 - T_1) - Q$$

$$x_A = \frac{C_p' (T_2 - T_1)}{-\Delta H_{R1} - (C_p'' - C_p') (T_2 - T_1)} = \frac{C_p' \Delta T}{-\Delta H_{R1} - (C_p'' - C_p') \Delta T}$$

---(2)

We have $\Delta H_{R2} = \Delta H_{R1} + (C_p'' - C_p') \Delta T$

$$x_A = \frac{C_p' \Delta T - Q}{-\Delta H_{R2}} = \frac{\text{Net heat still required after heat transfer to raise feed stream to } T_2}{\text{heat released by reaction at } T_2} \quad \text{---(3)}$$

When $C_p'' = C_p'$ the heat of re^n is independent of temp and we get,

$$x_A = \frac{C_p \Delta T - Q}{-\Delta H_R} \quad \text{---(4)}$$

With heat input proportional to ΔT , the energy balance line for adiabatic operation rotates about T_1 which is shown in fig.

Heat removal

Prop to

$$T_2 - T_1$$

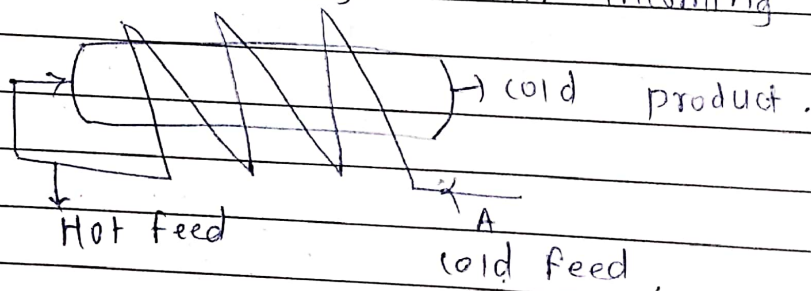
Energy balance line for
adiabatic operations

Heat addition

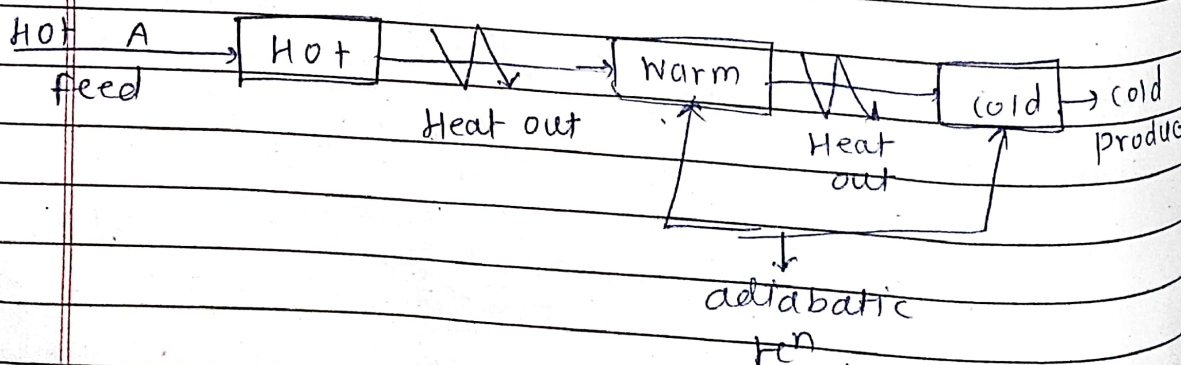
proportional to $T_2 - T_1$ T_1 = temp of feed. $T \rightarrow$

In case of exothermic rxn carried out in adiabatic manner, temp increase with conversion. Such cases, following schemes may be proposed to heat in order to achieve ideal temp profile.

- ① Heat exchange with incoming feed.



- ② multistage operation with interstage cooling between adiabatic sections.



Q. Explain the effect of temp on rate of rxn by following theories.

- i) collision theory
- 2) transition state theory.

→ (i) collision theory: The collision theory states

- that when suitable particles of a reactant hit each other, only a certain fraction of the collisions cause any noticeable or significant chemical change; these successful changes are called successful collisions.
- The successful collisions must have enough energy also known as activation energy, at the moment of impact to break the preexisting bonds and form all new bonds.
- This results in the products of the reaction increasing the concentration of the reactant particles or raising the temp - which brings about more collisions and hence more successful collisions therefore increase the rate of a rxn.
- When a catalyst is involved in the collision between the reactant molecule, less energy is required for the chemical change to take place, and hence more collisions have sufficient energy for reaction to occur the reaction rate therefore increases.

(2) Transition state theory

- transition state theory explains the reaction rates of elementary chemical reactions. the theory assumes a special type of chemical equilibrium (quasi-equilibrium) between reactants and activated transition state complex.
- TST is used primarily to understand qualitatively how chemical reactions take place.

TST has been less successful in its original goal of calculating absolute reaction rate constants because the calculations of absolute reaction rates requires precise knowledge of potential energy surfaces. However, it has been successful in calculating the standard enthalpy of activation ($\Delta H^\ddagger$) also written as $\Delta^\ddagger H^\ominus$, the standard entropy of activation $\Delta S^\ddagger$ and the standard Gibbs energy of activation $\Delta G^\ddagger$ for particular reaction if its rate constant has been experimentally determined.